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Genetic analysis of Preserved Ratio Impaired Spirometry using a population-based cohort

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Background

Preserved Ratio Impaired Spirometry (PRISm), defined as Forced Expiratory Volume in one second (FEV_1) <80% predicted with FEV_1/Forced Vital Capacity (FVC) ratio ≥0.70, is associated with respiratory symptoms, extra-pulmonary co-morbidities, and increased mortality.



Figure 1. Factors associated with PRISm including female sex, older age, smoking, comorbidities like diabetes and cardiovascular diseases, and genetics.

PRISm is a spirometric pattern that does not meet the criteria of obstructive lung disease, but may precede Chronic Obstructive Pulmonary Disease (COPD). PRISm resembles COPD in terms of FEV_1 <80% predicted, but not in FEV_1/FVC ratio.

Table1. Identifying spirometry measure patterns

Spirometry measures	Normal	PRISm	Obstruction
FEV_1	>80% predicted	<80% predicted	>or<80% predicted
FVC	>80% predicted	>80% predicted	>80% predicted
FEV_1/FVC	>70%	≥0.70	<70%

This study aims to understand the genetic architecture of PRISm in relation to other respiratory conditions, which may inform potential biological pathways and provide insight into the antecedents of irreversible lung damage.

Methods

Stage 1 – Discovery GWAS

We performed a genome-wide association study (GWAS) and determined genetic variants associated with PRISm using the UK Biobank, a large UK population-based health resource of about 500,000 people aged 40 to 69 at recruitment between 2006 and 2010.

Stage 2 – Replication Analysis

A replication GWAS was performed using the Extended Cohort for E-health, Environment and DNA (EXCEED), an on-going longitudinal population based cohort in Leicester, Leicestershire, and Rutland of over 10,000 participants aged 30-69 years old enrolled since the beginning of 2013.

Genetic correlation analysis

We utilised GWAS summary statistics from UK Biobank to assess genetic overlap of PRISm and other respiratory traits via genetic correlation analysis (r_g) using bivariate linkage disequilibrium (LD)-score regression.

Details of genetic association testing

- European ancestry
- BOLT-LMM logistic regression adjusting for age, sex, body mass index (BMI), smoking status, and principal components of genetic ancestry
- Population stratification was assessed using LD-score regression intercept
- genome-wide significance set at $p<5x10^{-8}$

Results

GWAS in UK Biobank

In a GWAS of PRISm in UK Biobank (38,639 cases and 257,643 controls), 27 SNPs from 16 chromosomes reached genome-wide significance ($p<5x10^{-8}$),

Replication analysis in EXCEED of top SNPs identified in UK Biobank GWAS

None of the 27 SNPs met a Bonferroni-corrected threshold ($p<0.002$) for independent replication in EXCEED (256 cases and 803 controls). Additional replication is taking place in further cohorts.

Genetic correlation analyses between PRISm and other respiratory traits

We found genetic correlations between PRISm and: childhood-onset asthma (r_g=0.131, p<0.001); adult-onset asthma (r_g=0.202, p<0.001); moderate-to-severe-asthma (r_g=0.310, p<0.001); asthma-Chronic Obstructive Pulmonary Disease (COPD) overlap syndrome (r_g=0.517, p<0.001) and; COPD (r_g=0.623, p<0.001). The magnitude of genetic correlation with PRISm corresponds to the degree of fixed airflow obstruction in each of these traits. In addition, we found a positive genetic correlation between respiratory infections and PRISm (r_g=0.185, p=0.003), suggesting shared aetiology (Figure 2).

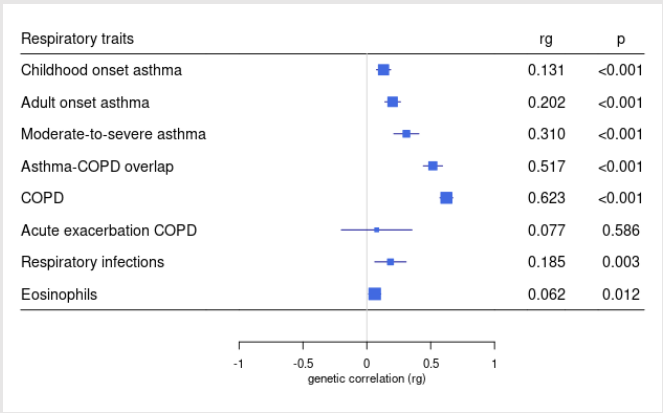


Figure 2. Forest plot showing the genetic correlation between PRISm and other respiratory traits. The genetic correlations of other respiratory traits GWAS were compared with UK Biobank PRISm GWAS.

Conclusion

- Our findings provide insights into the genetic architecture of PRISm in relation to other respiratory and complex diseases.
- 27 SNPs associated with PRISm in UK Biobank were not replicated in EXCEED potentially due to low power of the EXCEED GWAS.
- Respiratory diseases are genetically heterogeneous and share genetic risk factors with PRISm.
- A two-stage GWAS meta-analysis of PRISm, combining results of UK Biobank, EXCEED, and other studies is ongoing.

References

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